

## Lab 2 - Configuring Basic Single-Area OSPFv2

### Answer Sheet

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### Part 2: Configure and Verify OSPF Routing

#### Step 1: Verify OSPF neighbors and routing information.

b.

What command would you use to only see the OSPF routes in the routing table?

show ip route ospf

### Part 4: Configure OSPF Passive Interfaces

- g. Re-issue the **show ip route** and **show ip ospf neighbor** commands on R1 and R3, and look for a route to the 192.168.2.0/24 network.

What interface is R3 using to route to the 192.168.2.0/24 network? Serial0/0/0

What is the accumulated cost metric for the 192.168.2.0/24 network on R3? [110/120]

Does R2 show up as an OSPF neighbor on R1? yes

Does R2 show up as an OSPF neighbor on R3? no

What does this information tell you?

Traffic to the 192.168.2.0/24 network from R3 will be routed through R1, the S0/0/1 interface on R2 is still configured as a passive interface, so OSPF routing information is not being advertised across this interface.

- h. Change interface S0/0/1 on R2 to allow it to advertise OSPF routes. Record the commands used below.

router ospf 1

no passive-interface s0/0/1

- i. Re-issue the **show ip route** command on R3.

What interface is R3 using to route to the 192.168.2.0/24 network? Serial0/0/1

What is the accumulated cost metric for the 192.168.2.0/24 network on R3 now and how is this calculated?

[110/65]

Is R2 listed as an OSPF neighbor to R3? yes

## Part 5: Change OSPF Metrics

### Step 1: Change the reference bandwidth on the routers.

- a. To reset the reference-bandwidth back to its default value, issue the **auto-cost reference-bandwidth 100** command on all three routers.

```
R1(config)# router ospf 1
```

```
R1(config-router)# auto-cost reference-bandwidth 100
```

```
% OSPF: Reference bandwidth is changed.
```

Please ensure reference bandwidth is consistent across all routers.

Why would you want to change the OSPF default reference-bandwidth?

To obtain a more accurate cost calculation for these faster links, we need to set a higher default reference-bandwidth.

### Step 2: Change the bandwidth for an interface.

g.

Explain how the costs to the 192.168.3.0/24 and 192.168.23.0/30 networks from R1 were calculated.

The cost to 192.168.3.0/24 = 782 = 781 (R1 to R3) + 1 (R3 to the network)

The cost to 192.168.23.0/24 = 845 = 781 (R1 to R3) + 64 (R3 to the network)

- i. Issue the **bandwidth 128** command on all remaining serial interfaces in the topology.

What is the new accumulated cost to the 192.168.23.0/24 network on R1? Why?

1562

### Step 3: Change the route cost.

c.

Explain why the route to the 192.168.3.0/24 network on R1 is now going through R2?

Because the accumulated cost is lower (1563 < 1565)

1563 < 1565

### Reflection

1. Why is it important to control the router ID assignment when using the OSPF protocol?

Because the router ID in OSPF uniquely identifies a router in the OSPF domain. If we do not control the router ID, it may be assigned automatically, which can lead to inconsistent results. 4

By controlling the router ID, we can ensure stability and prevent unexpected changes, particularly in OSPF topology and route calculations.

2. Why is the DR/BDR election process not a concern in this lab?

Because we are using point-to-point links, and the DR/BDR election process does not apply. 4

3. Why would you want to set an OSPF interface to passive?

Because it could eliminate unnecessary OSPF routing information on that interface, freeing up bandwidth, the router will still advertise the network to its neighbours. 4

4. What aspect of OSPF makes the protocol hierarchical and permits the creation of very scalable networks?

Areas make the protocol hierarchical. By dividing a network into areas, OSPF reduces the size of the routing table and limits the scope of LSAs. This allows for more scalable networks with more routers and reduced network traffic. 4

5. What single OSPF router configuration command allows the assignment of OSPF area 0 to all interfaces in the range 10.0.0.0 to 10.255.255.255?

network 10.0.0.0 0.255.255.255 area 0 4

6. A router has three routes to the same network: one route from a RIP with a metric of 4, another from OSPF with a metric of 3053092, and another from EIGRP with a metric of 4039043. Which of the three routes will be used for the routing decision?

EIGRP route will be chosen because it has the lowest administrative distance. 4